An extension of convolutional neural networks to irregular domains through graph inference and translations identification

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I. Introduction

Convolutional neural networks (CNNs) have proven very efficient in tasks such as classification of images [1]. In such networks, the connectivity pattern of a convolutional layer — thus the number of weights to train — is reduced compared to a dense layer. The idea is to take advantage of the locality of objects to classify in images — *i.e.*, their localization on adjacent pixels — by linking neurons in a convolutional layer to those of the previous layer associated with close pixels only. Additionally, when considering natural scenes, images can typically be considered stationary signals. Thus, weights to train in a convolutional layer are shared among some connections of the layer, in order to detect a same object localized at various places of an image. This weight sharing is made by associating weights to train to a localized convolutional kernel that is then translated everywhere on the image. Any pixel covered by the kernel is then linked to the pixel associated with the kernel center in a resulting convolutional layer. The weight associated with this connection is the one associated with the kernel entry covering this first pixel.

The goal of this presentation is to introduce a framework in which we generalize this approach to signals evolving on an irregular, unknown graph. On such topology, there is no underlying metric space, and translation of a convolutional kernel as for images is not straightforward. We propose to proceed as described in Figure 1:

1) Infer a graph with well-chosen properties from a set of signals to classify;
2) Identify translations on this graph so that a signal retains its aspect when translated;
3) Use these translations to locate a convolutional kernel everywhere on the graph, thus defining the connectivity pattern of a convolutional layer;
4) Use these layers to build a CNN, and train it with the signals from which the whole process started.

In the next section, we briefly present the main ideas of points 1) and 2) above. Creation of the convolutional layer in 3) is done as described for images, using the approach in [2].

Compared to existing methods that generalize CNNs to irregular data, we exploit the inferred translations to obtain the weight sharing. Existing methods either work in the spectral domain (*e.g.*, [3]), or instantiate multiple kernels in the graph domain (*e.g.*, [4]). They then identify a mapping between localized kernels heuristically to simulate weight sharing.

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II. The introduced methodology

Our method first requires to infer a graph from data. Numerous methods exist depending on the assumptions made. Among possible assumptions, a smoothness prior [5] enforces similar vertices to be linked in the inferred graph, thus enforcing locality of objects in signals on this graph. Additionally, assuming stationarity of signals on the graph [6] implies some regularity in the inferred topology, allowing an object in a signal to look similar at various locations. These two assumptions can be combined [7] to obtain a resulting graph with interesting properties regarding our objective of mimicking CNNs on images for more complex data.

Translations on the inferred graph are not straightforward to identify due to the lack of a metric space. In [8], the authors have introduced translations that preserve the shape of translated objects. In details, such translations on a graph $\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle$ are functions $\psi : \mathcal{V} \mapsto \mathcal{V} \cup \{\perp\}$ such that:

- $\forall v_1, v_2 \in \mathcal{V} : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$, *i.e.*, ψ restricted to the vertices with an image in $\mathcal{V}$ is injective;
- $\forall v \in \mathcal{V} : (v, \psi(v)) \in \mathcal{E} \cup \psi(v) = \perp$, *i.e.*, vertices are sent to one of their neighbors;
- $\forall v_1, v_2 \in \mathcal{V} : (v_1, v_2) \in \mathcal{E} \Leftrightarrow (\psi(v_1), \psi(v_2)) \in \mathcal{E} \cup \psi(v_1) = \perp \cup \psi(v_2) = \perp$, *i.e.*, existing neighborhoods are preserved and no new neighborhoods are created.

In these criteria, a symbol $\perp$ is used to allow the *loss* of signal entries, due to border effects. Interestingly, the translations minimizing the loss (in the sense of the number of vertices having their image being $\perp$) on the grid graph are those that follow the *horizontal* and *vertical* directions. Therefore, in the case of irregular graphs, we expect translations minimizing the loss to be interesting for our usage, provided that irregularities are not too strong. However, identification of such translations is an NP-complete problem [9], and approximate translations should be found, for instance by exploiting the locality of the convolutional kernel. This will be detailed during the presentation associated with this summary.

III. Results

In order to show usefulness of our model, we have chosen to consider two datasets:

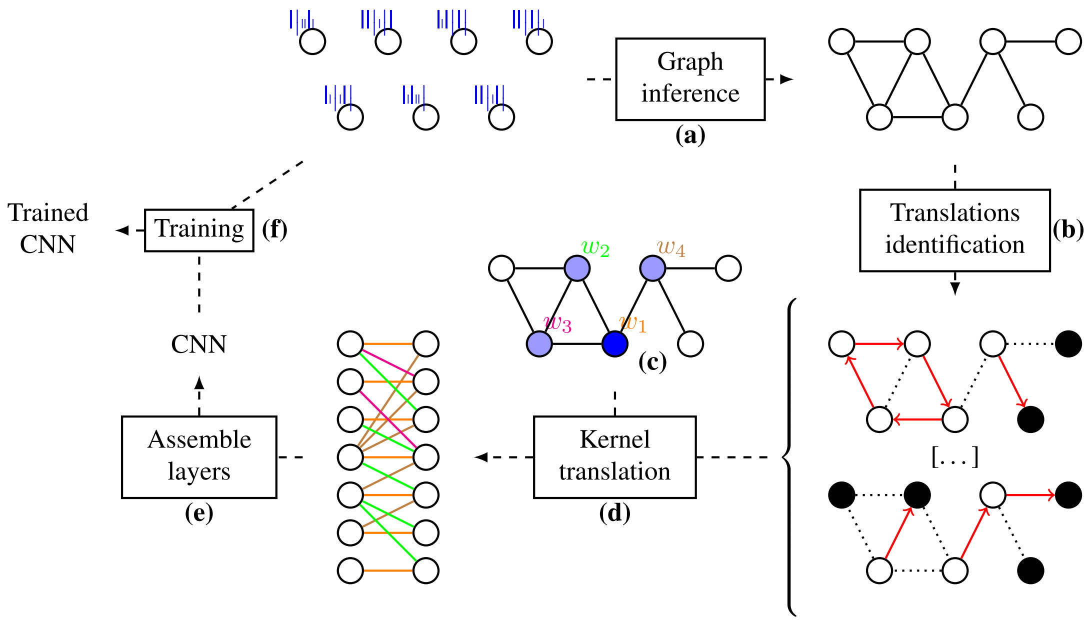

Figure 1: Introduced methodology to create a CNN from a given set of signals to classify. **(a)** From a set of signals observed on N variables, we infer a graph using some well-chosen assumptions. Here, we consider properties of smoothness (enforcing locality of objects) and stationarity (promoting regularity of the domain) of the signals on the graph; **(b)** We identify some translation operators on the graph. Note that this is an NP-complete problem. Therefore, approximate translations can be used here, found for example with greedy approaches, or exploiting the locality property of the kernel to translate; **(c)** We initialize a convolutional kernel on the graph, defined as a subset of close vertices associated with weights $(w_i)_i$ to train. Here, we choose to consider the most central vertex (in blue) as the kernel center, and to include its direct neighborhood (in mauve) in the kernel vertices; **(d)** Using the previously found translations, we translate the convolutional kernel so that it gets centered on every possible vertex. Once the kernel has been localized on every vertex, we can define a corresponding convolutional layer in the manner of [2]. Mapping of the various weights, represented by the colors in the resulting layer, is given directly by the translation of the initial kernel; **(e)** By considering various initializations of kernels, we obtain different convolutional layer models. A CNN can then be constructed by assembling them to form a complex network; **(f)** Finally, the CNN is trained using the data from which the whole process started. The trained network can then be used in order to classify new signals.

- A scrambled version of a dataset of images. This gives us a baseline to compare with state-of-the-art classification methods that do not use prior information that the signals are 2D images;
- A dataset of functional magnetic resonance imaging (fMRI) signals. This assesses applicability of our method to real-life signals.

The results we obtain on the scrambled images are very encouraging. Using smoothness and stationarity assumptions allows the reconstruction of a graph that is close to a grid. Translations then resemble Euclidean translations on images, which allows us to significantly outperform state-of-the-art methods that do not use prior on the topology of the signals.

The fMRI dataset however is more challenging, due to the presence of hubs in the inferred graph. This causes complexity issues, even when using a relaxed definition of the proposed translations. Using a different graph, for instance by linking close voxels in the brain, we have obtained results that outperform classifiers such as SVMs or linear regressions, but are comparable with a simple multi-layer perceptron. This is explained by the fact that such a geometric graph does not enforce locality of a pattern in the signal that may be representative of a particular class, thus preventing the CNN to take advantage of this.

References

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems*, 2012, pp. 1097–1105.

[2] J. Vialatte, V. Gripon, and G. Mercier, "Generalizing the convolution operator to extend cnns to irregular domains," *arXiv:1606.01166*, 2016.

[3] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Advances in Neural Information Processing Systems*, 2016, pp. 3844–3852.

[4] J. Atwood and D. Towsley, "Diffusion-convolutional neural networks," *arXiv:1511.02136*, 2015.

[5] V. Kalofolias, "How to learn a graph from smooth signals," in *Artificial Intelligence and Statistics*, 2016, pp. 920–929.

[6] B. Girault, "Stationary Graph Signals using an Isometric Graph Translation," in *Eusipco*, Nice, France, Aug. 2015, pp. 1531–1535. [Online]. Available: https://hal.inria.fr/hal-01155902

[7] B. Pasdeloup, V. Gripon, G. Mercier, D. Pastor, and M. G. Rabbat, "Characterization and inference of graph diffusion processes from observations of stationary signals," *IEEE Transactions on Signal and Information Processing over Networks*, 2017.

[8] N. Grelier, B. Pasdeloup, J.-C. Vialatte, and V. Gripon, "Neighborhood-preserving translations on graphs," in *GlobalSIP 2016 : IEEE Global Conference on Signal and Information Processing*, Arlington, United States, Dec 2016, pp. 410–414.

[9] B. Pasdeloup, "Extending convolutional neural networks to irregular domains through graph inference," Ph.D. dissertation, Ecole nationale supérieure Mines-Télécom Atlantique Bretagne Pays de la Loire, 2017.